

19	A	From the ideal gas equation: $pV = NkT$(1) and total internal energy of the gas molecules in the box, $U = \frac{3}{2}NkT$ Introducing additional N molecules of the same gas into the box $\Rightarrow p'V = (2N)kT'$(2) and total internal energy of the gas molecules, $U' = \frac{3}{2}(2N)kT'$, where p' & T' are the new pressure and temperature, and U' is the new internal energy of the gas in the box. $U = U' \rightarrow \frac{3}{2}NkT = \frac{3}{2}(2N)kT'$, resulting in $T' = \frac{1}{2}T$ $(2) \div (1) \rightarrow p' = p$
20	C	